
Samuele Scaglioni

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Experience

European Central Bank <i>Analyst</i>	Frankfurt, Germany <i>Oct 2025 – Present</i>
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- Contributing to SupTech Analytics workstreams applying AI and machine learning methods to supervisory data analysis, analytical tooling, and model development
- Co-developed a data strategy for the Directorate-General with the Economic Advisor, setting priorities on governance, architecture, and analytics enablement

European Central Bank <i>Stress Test Modelling Trainee</i>	Frankfurt, Germany <i>Oct 2024 – Sep 2025</i>
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- Designed a Hill-estimator tail-risk indicator with statistically significant predictive coefficients ($p < 0.05$) at 12–24-month horizons for model validation in the 2025 EU-wide stress test
- Developed a MATLAB pipeline and interactive dashboard to automate analysis of model questionnaires from 80+ Euro Area banks for the 2025 EU-wide stress test, cutting review time by **~50%**

Research Experience

European Central Bank <i>Researcher, Financial Stability and Stress Testing</i>	Frankfurt, Germany <i>Oct 2024 – Present</i>
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- Working paper on probability of default and micro-level stress testing, applying machine learning methods to bank-level credit risk modelling and stress testing
- Research assistant to Dr. Aurea Marques, Credit Risk Team Lead, on trade wars, tariff shocks, and abnormal returns using event-study methods and GJR–GARCH models

Politecnico di Milano <i>Research Assistant, Macroeconomics</i>	Milan, Italy <i>Nov 2024 – Nov 2025</i>
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- With Prof. Daniele Siena, analyzed ECB conventional and unconventional policy shocks using daily yield-curve movements in a Proxy SVAR with functional-PCA identification.

University of Illinois Chicago <i>Research Assistant, Data Science and Econometrics</i>	Chicago, USA <i>Jan 2024 – Jun 2024</i>
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- Co-developed *IMPACT*, a causal-ML pipeline using random forests and causal forests for CATE estimation and budget-constrained targeting, presented at the 2024 INFORMS Annual Meeting.

Working Papers

Learning Probability of Default and Stress Testing

Nocciola, L. and S. Scaglioni. 2026. *ECB Working Paper Series*, No. XXXX.

- Presented at the EcoMod2026 International Conference on Economic Modeling and Data Science, Luxembourg, July, 2026, and at the 32nd International Conference on Computing in Economics and Finance (CEF 2026), Ca' Foscari University of Venice, July 1, 2026

Institutional Cleansing and Political Scars: Mafia-Related Municipal Dissolution, Electoral Participation, Political Supply, and Credit Disruption in Italy

Abidi, N., I. Miquel-Flores, and S. Scaglioni. 2026. *ECB Working Paper Series*, No. XXXX.

- Shows that mafia-related municipal dissolution leaves a persistent political participation scar, disrupts local political supply, and is associated with short-run credit-market spillovers

Trade Wars, Tariff Shocks, and Abnormal Returns: Evidence from a GJR–GARCH Event Study

Mimeo, 2025, with Aurea Marques (European Central Bank), Agha, Durrani (European Central Bank)

- Documents statistically significant abnormal returns around tariff announcements, most pronounced in import-intensive sectors, with asymmetric volatility captured by GJR–GARCH and stronger downside responses

IMPACT: A framework for robust causal treatment analysis in resource-constrained settings

Paper Submitted to EAAI, 2025, with Hadis Anahideh (University of Illinois at Chicago)

- Delivers a budget-aware CATE framework that ranks individuals for treatment, simulates counterfactual outcomes, and improves policy targeting—outperforming propensity-score baselines out of sample

Education

University of Chicago

Master of Arts in Economics

Chicago, USA

Sep 2026 – Jun 2027

- Dean's Scholarship

University of Illinois Chicago

MSc in Industrial Engineering

Chicago, USA

Jan 2023 – Dec 2024

- GPA: 4.0/4.0 (top 1%)

Politecnico di Milano

MSc in Management and Production Engineering

Milan, Italy

Sep 2022 – Jul 2024

- GPA: 29/30; Final grade: 110/110 *cum laude* (top 1%)

Politecnico di Milano

BSc in Management and Production Engineering

Milan, Italy

Sep 2018 – Jul 2022

- GPA: 25/30

- Erasmus + scholarship (€5000), exchange at **South-Eastern Finland University of Applied Sciences**

Math & Economics Coursework (selected)

Math (proof-based): Mathematical Analysis I, II (equivalent of real analysis) [UG]; Linear Algebra & Geometry [UG]; Probability & Statistics [UG]; Mathematical Methods for Operations Research and Game Theory [UG]

Stochastic/Time Series: Stochastic Processes [Grad]; Quality Data Analysis [Grad]

Econometrics & Data: Data Science 1 [Grad]; Business Data Analytics [UG]

Economics (calculus-based): Microeconomics [UG]; Macroeconomics [UG]

Economics (theory): Economics for Development [Grad]; Economics of Multinational Enterprises [Grad]

Other Honors & Awards

GRE: 165 Quantitative, 165 Verbal, 4.0 AWA GMAT equivalent: 720 ($\approx$ 95th percentile)

50% tuition waiver, University of Illinois Chicago

Member (2023)—**NOVA**: selective global network of top 3% high-achieving individuals

Relevant Experience

ISPI — Istituto per gli Studi di Politica Internazionale

Diploma in Geopolitics & Global Studies

Milan, Italy

202–202

- Undergraduate-level program covering geopolitics, geo-economics, and international security.
- Curriculum highlights: Theories and Rules of International Trade; New Challenges for Global Security in the 21st Century; NATO—Transformation, Thoughts, and Future Dilemmas; Russia's Geopolitical Vision—A Global Dimension; Politics and Energy in the Mediterranean.

Mount Albert Grammar School

High School Exchange Student & Fencing Team Captain

Auckland, New Zealand

Jul 2017 – Sep 2017

- Captain of the foil fencing team; placed 3rd in the New Zealand U-18 National Foil Competition.

Languages, Skills & Interests

Languages: English (C2 Cambridge, IELTS 8.5), Italian (Native)

Computer Science: Advanced in Python (MIT Certificate), MATLAB and Excel; Intermediate in SQL and LaTeX

Interests: Sled dogs Volunteering (Two months in Alberta, Canada, Jun–Jul 2018), Chess, Gym, Movies